UNNI ADITYA MOHAN

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EDUCATION

Master of Science in Robotics, Northeastern University, Boston, MA — Sep 2024 – Dec 2026

Bachelor of Technology in Mechanical Engineering, NIT Rourkela, India — Aug 2017 – Jun 2021

EXPERIENCE

Robotics Engineer, Black-I Robotics, Boston, MA — Sep 2025 – Dec 2025

- Pretrained **YOLOv8-seg** on **2,000** auto-labeled **Isaac Sim Replicator** pallet scenes, then fine-tuned on 600 real images.
- Built **C++** inference for box pose, long-edge yaw and size with 5-frame averaging and box-database cross-checks, feeding a **multi-box grouping algorithm** that outputs gripper x, y, z, yaw and a **3-zone suction mask**.
- Replaced the MoveIt planner with **cuRobo** trajectory optimization over a 6-camera nvblox ESDF behind a **200 Hz CBF-QP safety filter**, rebuilding the robot model from CAD (SolidWorks, STL, URDF, **USD**) with collision spheres.
- Redesigned the forearm with a 15 cm extension to move the arm off its singular full-stretch pose, validating stiffness, deflection and vibration in **ANSYS FEA** and specifying per-joint torque requirements.

Research Assistant, MuRAL, Northeastern University — Mar 2025 – Present

- Integrated and calibrated dual **load cells** and a payload angle sensor on an omnidirectional robot fleet, running a **20 Hz force-acquisition loop** and an admittance controller with obstacle-triggered avoidance.
- Co-authored an **ACC 2027** paper (submitted) on communication-free transport with a **piecewise-linear** twist-to-wrench model.
- Recovered obstacle bearing and distance from aggregate wrench measurements via active test-motion probing and **rank-one eigen-decomposition**, validated across **30 simulated scenes** with up to 10 followers.

Project Engineer, Thermax Ltd., Pune, India — Sep 2021 – May 2024

- Built a **surrogate model** for steam-drum thermal parameters, replacing design iterations and saving **20 engineering hours** per cycle.

PROJECTS

RL-Fine-Tuned VLA for Humanoid Manipulation from Egocentric Phone Data | *MuJoCo*, *LeRobot (SmolVLA)*, *MediaPipe*, *PyTorch* — Oct 2026

- Retargeted my own **egocentric phone videos** (MediaPipe hand tracking) to a **Unitree G1 + Dex3-1** in MuJoCo, yielding **584 contact-only** pick-and-place demos.
- Fine-tuned **SmolVLA** on task-frame hand-pose actions, then improved it with **advantage-weighted RL** on its own rollouts (37% to 45%).
- Traced failures to grasp precision and added a **wrist camera**, reaching **52%** on 60 unseen poses with **0 falls**; benchmarked vs an **RRT-Connect + IK** demo pipeline (73%).

Sim-to-Real Humanoid Locomotion, Unitree G1 (29-DoF) | *MuJoCo Warp*, *RSL-RL*, *PyTorch*, *ONNX* — Sep 2026 – Present

- Built a **parity harness** verifying the 50 Hz ONNX runtime reproduces training observations and actions to **3e-6** before hardware deployment.
- Traced a **hardware freeze** to zero joint friction and damping in the **MJCF**, fixing the sim model with friction, gain and latency randomization for **first physical locomotion**.
- Ran **chirp-excitation system ID** on 12 leg joints, cutting held-out sim-to-real RMSE **31%** and setting randomization ranges.
- Added tilt, staleness and NaN **fallbacks** to the onboard runner, deploying height-conditioned **crouch-walking** policies on hardware and training payload robustness to 10 kg in sim.
- Redesigned PPO training and rewards across **4,096 parallel MuJoCo Warp environments** on a single 8 GB RTX 4060.

GRASP, Language-Conditioned Sequential Manipulation | *MuJoCo*, *PyTorch*, *OWLv2* — Jul 2026 – Present

- Built a **52-check automated validation suite** for an architecture-agnostic MuJoCo sequential pick-and-place benchmark across 9 objects and 3 destination types.
- Stabilized **contact dynamics** with convex-decomposed colliders, elliptic friction cones, 500 Hz implicit integration, fingertip friction and a 6x stiffer gripper servo.
- Developed DLS IK with null-space posture control and an antipodal grasp planner yielding valid grasps for **98%** of objects.
- Grounded language commands in **simulated dual RGB-D camera feeds** with OWLv2 for camera-only perception, reaching **4.7 mm** mean 3D position error.

Trajectory Planning for UAV in Unknown Dynamic Environments | *ROS*, *Gazebo*, *Gurobi*, *Docker* — Feb 2026 – May 2026

- Built a **reproducible Dockerized Gazebo benchmark**, reaching **9/9 vs 6/9** success over vanilla FASTER with 8 obstacles at 0.5–1.5 m/s in a 70 m corridor.
- Built a 3-frame differencing **velocity predictor** for the **FASTER** planner (IEEE T-RO) that creates, shrinks and re-creates speed-adaptive safety zones around moving obstacles, keeping the original **MIQP** unchanged.

TECHNICAL SKILLS

Software & Tools: C++, Python, PyTorch (CUDA), ROS / ROS 2, ONNX, Docker, Git, Linux, SolidWorks, ANSYS FEA
Simulation & Validation: Isaac Sim (Replicator), MuJoCo (Warp), Gazebo, USD, URDF, MJCF, Domain Randomization, Contact Tuning, System ID, Sim-to-Real Parity Testing
Learning & Perception: RL (PPO, SAC, AWR), VLA Fine-Tuning (SmolVLA), YOLOv8-seg, OWLv2, MediaPipe, nvblox, SLAM (LIO-SAM, GTSAM), OpenCV
Controls & Planning: Trajectory Optimization (QP, MIQP), cuRobo, CBF Safety Filters, MPC, Admittance Control, Inverse Kinematics